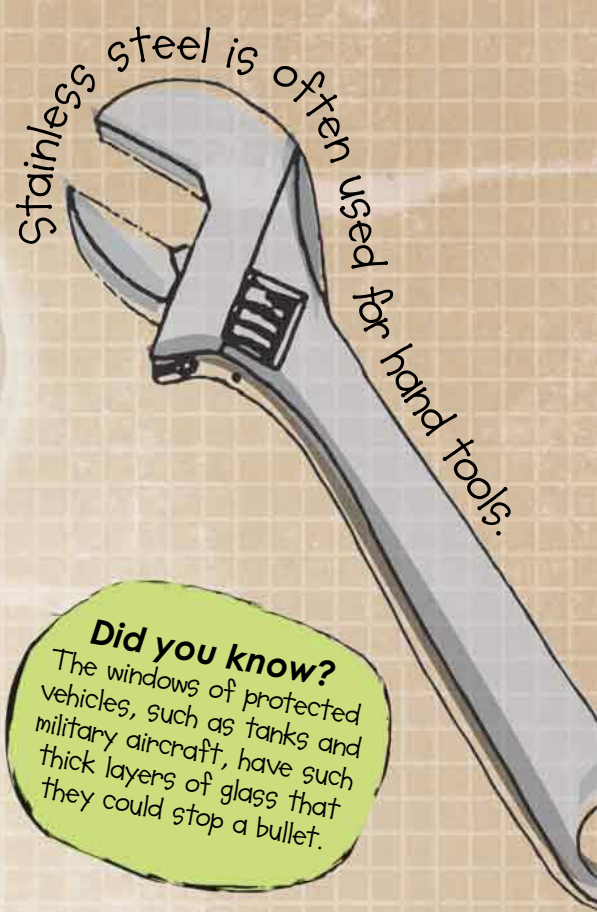
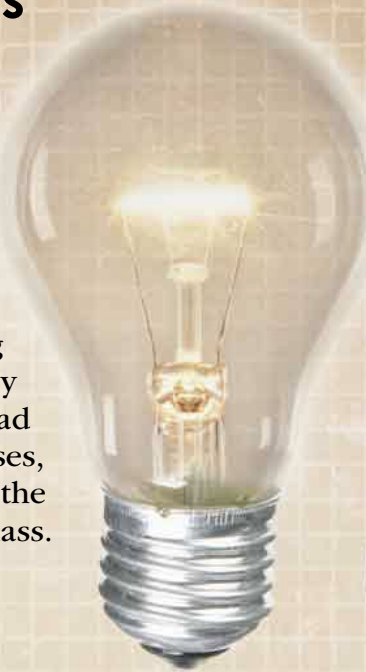


Glass

Heating the mineral sodium carbonate and sand produces glass, but it is unclear who discovered the process. The Egyptians were **glazing jewelry beads** in 2500 BCE, while the ancient Romans were the first to use transparent glass. Around **1000 CE**, the glass industry took off in Europe, with **THE VENETIANS OF ITALY** producing the finest glass for centuries afterward. By the 17th century, most Europeans had glass **windowpanes** at home. Eyeglasses, mirrors, and light bulbs are among the many inventions that make use of glass.



Did you know?
The windows of protected vehicles, such as tanks and military aircraft, have such thick layers of glass that they could stop a bullet.

Kevlar

This **synthetic fiber** is a relatively young invention in the material world. Created by American chemist **STEPHANIE KWOLEK** and patented in 1966, Kevlar is five times stronger than steel. In addition to its **super strength**, it is lightweight and does not rust. This makes the tightly woven fibers ideal for the **BULLETPROOF VESTS** worn by police officers. When it's not saving lives, Kevlar is used to make parachutes, skis, cell phones, construction clothing, bicycle tires, and **underwater cables**.



Kevlar makes these work gloves cut-proof.

Titanium

In 1791, English clergyman **William Gregor** discovered titanium in its mineral form, but it wasn't until 1932 that Luxembourg metal expert William Kroll produced the **METALLIC TITANIUM** used today. Removing impurities results in a material **as strong as steel but half as heavy**. Named after the Titans of Greek mythology, titanium is used in aircraft, spacecraft, boats, bicycles, and machine parts like these cogs.

